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Федеральное государственное бюджетное образовательное
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РТУ МИРЭА
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Кафедра КБ-4 «Интеллектуальные системы информационной
безопасности»

Отчет по практической работе №3
по дисциплине «Системы мониторинга и управления инцидентами
информационной безопасности»

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Характеристика	Snort (v2.9 / v3)	Suricata	Zeek (ранее Bro)
Тип системы	Сигнатурная IDS/IPS	Сигнатурная IDS/IPS	Аналитическая платформа (NSM)
Модель обработки	Однопоточная (v2), Многopоточная (v3)	Многopоточная (Native)	Событийно-ориентированная
Язык разработки	C / C++	C, Rust (в новых версиях)	C++ (ядро), Zeek Script
Метод детекции	Сопоставление с паттернами	Сопоставление с паттернами	Поведенческий анализ / Скрипты
Основной формат логов	Unified2, Syslog	JSON (Eve box), PCAP	Табулированный текст, JSON
Уровень OSI	Snort	Suricata	Zeek
L2 (Канальный)	Базовый разбор заголовков	Поддержка VLAN, MPLS	Глубокий разбор инкапсуляций
L3/L4 (Сетевой)	Стандартная фильтрация IP/Порт	Высокая производительность потоков	Хранение состояний (Stateful)
L7 (Прикладной)	Ограничен (через препроцессоры)	Глубокая инспекция (DPI), извлечение файлов	Максимальная детализация, скриптовый разбор протоколов
Протоколы	HTTP, FTP, SMTP	HTTP, TLS (с разбором сертов), DNS	Любые (возможность написания парсера на Spicy)

Система	Расположение конфигов (типичное)	Язык правил / логики	Особенности синтаксиса
Snort	/etc/snort/snort.conf	Собственные (Rule-based)	<pre> alert tcp \$EXTERNAL_NET any -> \$HTTP_SERVERS 80 </pre>
Suricata	/etc/suricata/suricata.yaml	YAML + Snort-compatible rules	Поддержка расширенных ключевых слов (lua-скрипты)
Zeek	/opt/zeek/share/zeek/site/local.zeek	Zeek Script (Тьюринг-полный)	<pre> event http_request(...) { ... } — логика как в программировании </pre>

Корреляция для вазуха

```

<localfile>
  <log_format>json</log_format>
  <location>/var/log/snort/alert_json.txt</location>
</localfile>

<localfile>
  <log_format>json</log_format>
  <location>/var/log/suricata/eve.json</location>
</localfile>

<localfile>
  <log_format>syslog</log_format>
  <location>/opt/zeek/logs/current/http.log</location>
</localfile>

```

```

GNU nano 8.4 /var/ossec/etc/rules/local_rules.xml *
<group name="snort_json,web,accesslog,">

```

```

<rule id="100010" level="3">
  <decoded_as>json</decoded_as>
  <field name="msg">\.+</field>
  <description>Snort Event: $(msg)</description>
  <options>no_full_log</options>
</rule>

<rule id="100012" level="12">
  <if_sid>100010</if_sid>
  <field name="msg">Potential TCP SYN Flood</field>
  <description>Critical Attack: TCP SYN Flood detected via Snort
(Level 12)</description>
</rule>

<rule id="100030" level="12">
  <if_sid>31100</if_sid>
  <match>UNION SELECT|database()</match>
  <description>Critical Attack: SQL Injection attempt via Apache (Level
12)</description>
</rule>

<rule id="100100" level="1">
  <if_sid>31100</if_sid>
  <description>Apache Request log</description>
</rule>

<rule id="100200" level="12" frequency="80" timeframe="20">
  <if_matched_sid>100100</if_matched_sid>
  <description>Critical Attack: HTTP DDoS detected via Wazuh
Correlation (Level 12)</description>
  <group>attack,pci_dss_11.4,</group>
</rule>

</group>

<group name="zeek_alerts,">

<rule id="100050" level="0">

```

```
<decoded_as>syslog</decoded_as>

<location>/opt/zeek/logs/current/http.log</location>

<description>Zeek HTTP log event</description>

</rule>


<rule id="100051" level="12">

  <if_sid>100050</if_sid>

  <match>UNION SELECT|UNION ALL SELECT</match>

  <description>Zeek: Detected SQL Injection attempt in HTTP
URI</description>

  <group>attack,sql_injection,pci_dss_11.4,</group>


</rule>


<rule id="100052" level="12">

  <if_sid>100050</if_sid>

  <match>\<script>|alert\(|%3Cscript%3E</match>

  <description>Zeek: Detected XSS attempt in HTTP URI</description>

  <group>attack,xss,</group>

</rule>

</group>
```

Установка суриката

```
user@debian:~$ sudo apt update && sudo apt install suricata -y
Get:1 http://security.debian.org/debian-security trixie-security InRelease [43.4 kB]
Get:2 http://deb.debian.org/debian trixie InRelease [140 kB]
Get:3 http://deb.debian.org/debian trixie-updates InRelease [47.3 kB]
Get:4 http://security.debian.org/debian-security trixie-security/main Sources [126 kB]
Get:5 http://deb.debian.org/debian trixie/main Sources [10.5 MB]
Get:6 http://security.debian.org/debian-security trixie-security/main amd64 Packages [121 kB]
Get:7 http://security.debian.org/debian-security trixie-security/main Translation-en [76.0 kB]
Get:8 http://deb.debian.org/debian trixie/main amd64 Packages [9,671 kB]
Get:9 http://deb.debian.org/debian trixie/main Translation-en [6,484 kB]
Fetched 27.2 MB in 6s (4,697 kB/s)
50 packages can be upgraded. Run 'apt list --upgradable' to see them.
Notice: Repository 'http://deb.debian.org/debian trixie InRelease' changed its 'Version' value from '13.3' to '13.4'
Installing:
  suricata

Installing dependencies:
  isa-support          libnet1              librte-ip-frag25     libxdp1
  libevent-2.1-7t64   libnetfilter-log1    librte-kvargs25      libyaml-0-2
  libevent-core-2.1-7t64 libnetfilter-queue1  librte-log25         python3
  libevent-pthreads-2.1-7t64 libnfnetlink0        librte-mbuf25        python3-minimal
  libfdt1             libpcap0.8t64        librte-mempool25     python3-yaml
  libhiredis1.1.0     libpython3-stdlib    librte-meter25       python3.13
  libhttp2            libpython3.13-minimal librte-net-bond25     python3.13-minimal
  libhyperscan5       libpython3.13-stdlib librte-net25          sse3-support
  libluajit-5.1-2     librte-bus-pci25     librte-pci25         sse4.2-support
  libluajit-5.1-common librte-bus-vdev25    librte-rcu25         suricata-update
  libmagic-mgc        librte-eal25         librte-ring25
  libmagic1t64       librte-ethdev25      librte-sched25
  libmaxminddb0      librte-hash25        librte-telemetry25
```

Рисунок 1 – Установка Suricata

```
user@debian:~$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:96:bb:42 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx00c2996bb42
    inet 192.168.65.129/24 brd 192.168.65.255 scope global dynamic noprefixroute ens33
        valid_lft 1123sec preferred_lft 898sec
    inet6 fe80::13e7:2dde:1cf:ec83/64 scope link
        valid_lft forever preferred_lft forever
user@debian:~$
```

Рисунок 2 – Проверка порта

```

root@debian:/home/user# systemctl status suricata
● suricata.service - Suricata IDS/IDP daemon
   Loaded: loaded (/usr/lib/systemd/system/suricata.service; enabled; pre>
   Active: active (running) since Sat 2026-04-11 08:31:35 EDT; 15s ago
   Invocation: 31e29eable4e4761b16f1458c6e1c742
      Docs: man:suricata(8)
            man:suricatasc(8)
            https://suricata.io/documentation/
   Process: 73801 ExecStart=/usr/bin/suricata -D --af-packet -c /etc/suric>
   Main PID: 73802 (Suricata-Main)
      Tasks: 1 (limit: 2255)
     Memory: 480.4M (peak: 498.7M)
        CPU: 15.670s
       CGroup: /system.slice/suricata.service
              └─73802 /usr/bin/suricata -D --af-packet -c /etc/suricata/suri>

```

Рисунок 3 – Проверка статуса Suricata

Установка snort

```

root@debian:/# sudo docker run -it --rm --net=host ciscotalos/snort3 snort -
V
Unable to find image 'ciscotalos/snort3:latest' locally
latest: Pulling from ciscotalos/snort3
281b80c799de: Pull complete
b6471ccb36a8: Pull complete
007d993cceb3: Pull complete
a53b597c0ca5: Pull complete
48372e25bf7a: Pull complete
d486d7068f22: Pull complete
b8ff25e157c8: Pull complete
Digest: sha256:f15f714fd61ade114b1039687dd43352cae26cd6401b89f2b03404e0d21e3
22f
Status: Downloaded newer image for ciscotalos/snort3:latest
snorty@debian:~$ snort --version

o''~)~
'''

--> Snort++ <*-
Version 3.10.0.0
By Martin Roesch & The Snort Team
http://snort.org/contact#team
Copyright (C) 2014-2025 Cisco and/or its affiliates. All rights r
eserved.

Copyright (C) 1998-2013 Sourcefire, Inc., et al.
Using DAQ version 3.0.23
Using Hyperscan version 5.4.2 2024-12-30
Using LibML version 2.0.0
Using libpcap version 1.10.5 (with TPACKET_V3)
Using LuaJIT version 2.1.1737090214
Using LZMA version 5.8.1
Using OpenSSL 3.5.4 30 Sep 2025
Using PCRE2 version 10.46 2025-08-27
Using ZLIB version 1.3.1

snorty@debian:~$ |

```

Рисунок 4 – Установка Snort

```

user@debian:~$ sudo nano /usr/local/etc/snort/rules/local.rules

```

```

alert icmp $EXTERNAL_NET any -> $HOME_NET any (msg:"ICMP
Flood Attempt"; detection_filter:track by_src, count 20, seconds 10;
sid:1000001; rev:1;)

```



```
alert icmp $EXTERNAL_NET any -> $HOME_NET any (msg:"ICMP
Large Packet Detected"; dsize:>1000; itype:8; sid:1000002; rev:1;)
```

Установка Zeek

```
user@debian:~$
user@debian:~$ echo 'deb http://download.opensuse.org/repositories/security:/zeek/xUbuntu_24.04/ /' |
do tee /etc/apt/sources.list.d/security:zeek.list
deb http://download.opensuse.org/repositories/security:/zeek/xUbuntu_24.04/
user@debian:~$ curl -fsSL https://download.opensuse.org/repositories/security:zeek/xUbuntu_24.04/Release
.key | \
> gpg --dearmor | sudo tee /etc/apt/trusted.gpg.d/security_zeek.gpg > /dev/null
user@debian:~$ apt update -y
Notice: Ignoring file 'security:zeek.listecho' in directory '/etc/apt/sources.list.d/' as it has an i
lid filename extension
Error: Could not open lock file /var/lib/apt/lists/lock - open (13: Permission denied)
Error: Unable to lock directory /var/lib/apt/lists/
Warning: Problem unlinking the file /var/cache/apt/pkgcache.bin - RemoveCaches (13: Permission denied)
Warning: Problem unlinking the file /var/cache/apt/srcpkgcache.bin - RemoveCaches (13: Permission den
)
user@debian:~$ apt install zeek -y
Notice: Ignoring file 'security:zeek.listecho' in directory '/etc/apt/sources.list.d/' as it has an i
lid filename extension
Error: Could not open lock file /var/lib/dpkg/lock-frontent - open (13: Permission denied)
Error: Unable to acquire the dpkg frontend lock (/var/lib/dpkg/lock-frontent), are you root?
user@debian:~$ sudo apt update -y
Get:1 http://security.debian.org/debian-security trixie-security InRelease [43.4 kB]
Hit:2 http://deb.debian.org/debian trixie InRelease
Get:3 http://deb.debian.org/debian trixie-updates InRelease [47.3 kB]
Get:4 http://security.debian.org/debian-security trixie-security/main Sources [132 kB]
Get:5 http://download.opensuse.org/repositories/security:/zeek/xUbuntu_24.04 InRelease [1,937 B]
Get:6 http://download.opensuse.org/repositories/security:/zeek/xUbuntu_24.04 Packages [9,220 B]
Get:7 http://security.debian.org/debian-security trixie-security/main amd64 Packages [127 kB]
Get:8 http://security.debian.org/debian-security trixie-security/main Translation-en [80.0 kB]
Fetched 440 kB in 1s (376 kB/s)
47 packages can be upgraded. Run 'apt list --upgradable' to see them.
Notice: Ignoring file 'security:zeek.listecho' in directory '/etc/apt/sources.list.d/' as it has an i
lid filename extension
user@debian:~$ sudo apt install zeek -y
Installing:
  zeek

Installing dependencies:
libbroker-dev      python3-aiodns      python3-gitdb       sendmail-base
libcares2          python3-aiohappyeyeballs python3-idna         sendmail-bin
liblockfile-bin   python3-aiohttp     python3-multidict    sendmail-cf
liblockfile1      python3-aiosignal   python3-propcache    zeek-btest
libmaxminddb-dev  python3-aiosqlite   python3-pycares      zeek-btest-data
libnorm1t64       python3-async-timeout python3-redis         zeek-client
libnsl2           python3-asynccpg    python3-semantic-version zeek-core
libpgm-5.3-0t64   python3-attr        python3-smmmap        zeek-core-dev
libzmq5           python3-cffi-backend python3-typing-extensions zeek-spicy-dev
lockfile-progs    python3-frozenset   python3-uvloop        zeek-zkg
postfix-mta-sts-resolver python3-git          python3-yarl          zeekctl
```

Рисунок 5 – Установка Zeek

```
root@debian:/home/user# zeek --version
zeek version 8.1.1
root@debian:/home/user# |
```

Рисунок 6 – Проверка версии zeek

Правила для Zeek

```
@load base/protocols/http
@load base/frameworks/notice

module MySecurity;

export {
  redef enum Notice::Type += {
    SQL_Injection_Attempt,
    Possible_Flood_Attack
  };

  # Порог для обнаружения флуда (например, больше 50
  соединений от одного IP)
  const flood_threshold = 50;
}

SQL

event http_request(c: connection, method: string, original_URI: string,
unescaped_URI: string, version: string)
{
  # Приводим URI к нижнему регистру для поиска
  local uri = to_lower(unescaped_URI);

  if ( /union/ in uri && /select/ in uri || /select/ in uri && /from/ in uri )
  {
    NOTICE([$note=SQL_Injection_Attempt,
              $conn=c,
              $msg=fmt("Подозрение на SQLi в URI: %s",
unescaped_URI),
              $identifier=cat(c$Id$orig_h)]);
  }
}

global connection_count: table[addr] of count &default=0;

event new_connection(c: connection)
{
  local src_ip = c$Id$orig_h;
  connection_count[src_ip] += 1;

  if ( connection_count[src_ip] > flood_threshold )
```

```

{
    NOTICE([$note=Possible_Flood_Attack,
            $conn=c,
            $msg=fmt("Слишком много соединений от %s: %d", src_ip,
connection_count[src_ip]),
            $identifier=cat(src_ip)]);
    }
}

```

DDoS атака:

```

slasher@DESKTOP-0JC91UK:/mnt/c/Users/Dasha$ sudo hping3 -S --flood -V -p 80 192.168.65.129
using eth0, addr: 172.25.187.219, MTU: 1500
HPING 192.168.65.129 (eth0 192.168.65.129): S set, 40 headers + 0 data bytes
hping in flood mode, no replies will be shown

```

Рисунок 7 – Имитация ddos атаки

Suricata

data.timestamp	2026-03-31T15:37:27.328582-0400
decoder.name	json
id	1774985851.16440264
input.type	log
location	/var/log/suricata/eve.json
manager.name	debian
rule.description	Suricata: Alert - Suricata Attack: Potential TCP SYN Flood
rule.firedtimes	10
rule.groups	ids, suricata
rule.id	86601
rule.level	3
rule.mail	false

Рисунок 8 – Результат атаки на Suricata

Snort

t decoder.name	json
t id	1774978204.12845775
t input.type	log
t location	/var/log/snort/alert_json.txt
t manager.name	debian
t rule.description	Snort Attack: TCP SYN Flood detected
# rule.firedtimes	350
t rule.groups	snort_json
t rule.id	100012
# rule.level	10
rule.mail	false
timestamp	Mar 31, 2026 @ 20:30:04.108

Рисунок 9 – Результат атаки на Snort

Zeek

```

root@debian:/opt/zeek/bin# tail -f /opt/zeek/logs/current/notice.log
1774989379.878820 - - - - - MySecu
rity::DDoS_SYN_Flood Flood detected from 192.168.8.1 - 192.168.8.1 - - -
Notice::ACTION_LOG (empty) 3600.000000 - - - - -
1774989382.398568 - - - - - MySecu
rity::DDoS_SYN_Flood Flood detected from 192.168.8.1 - 192.168.8.1 - - -
Notice::ACTION_LOG (empty) 3600.000000 - - - - -
1774989389.479620 - - - - - MySecu
rity::DDoS_SYN_Flood Flood detected from 192.168.8.128 - 192.168.8.128 - - -
Notice::ACTION_LOG (empty) 3600.000000 - - - - -
1774989393.943777 - - - - - MySecu
rity::DDoS_SYN_Flood Flood detected from 192.168.8.1 - 192.168.8.1 - - -
Notice::ACTION_LOG (empty) 3600.000000 - - - - -
1774989398.228136 - - - - - MySecu
rity::DDoS_SYN_Flood Flood detected from 192.168.8.128 - 192.168.8.128 - - -
Notice::ACTION_LOG (empty) 3600.000000 - - - - -
1774989398.251952 - - - - - MySecu
rity::DDoS_SYN_Flood Flood detected from 192.168.8.128 - 192.168.8.128 - - -
Notice::ACTION_LOG (empty) 3600.000000 - - - - -
1774989398.256279 - - - - - MySecu
rity::DDoS_SYN_Flood Flood detected from 192.168.8.128 - 192.168.8.128 - - -
Notice::ACTION_LOG (empty) 3600.000000 - - - - -
1774989409.214832 - - - - - MySecu
rity::DDoS_SYN_Flood Flood detected from 192.168.8.1 - 192.168.8.1 - - -
Notice::ACTION_LOG (empty) 3600.000000 - - - - -
1774989421.135123 - - - - - MySecu
rity::DDoS_SYN_Flood Flood detected from 192.168.8.1 - 192.168.8.1 - - -
Notice::ACTION_LOG (empty) 3600.000000 - - - - -
1774989431.340993 - - - - - Captur
eLoss::Too_Little_Traffic Only observed 0 TCP ACKs and was expecting at least 1. - - -
Notice::ACTION_LOG (empty) 3600.000000 - - - - -

```

Рисунок 10 – Результат атаки на Zeek

SQL-injection

Suricata

Time	_source
> Mar 31, 2026 @ 22:40:21.785	input.type: log agent.ip: 192.168.8.128 agent.name: Agent_1 agent.id: 004 manager.name: debian data.tx_id: 0 data.app_proto: http data.in_iface: ens33 data.src_ip: 192.168.8.1 data.src_port: 62261 data.event_type: alert data.alert.severity: 3 data.alert.signature_id: 1000004 data.alert.rev: 1 data.alert.gid: 1 data.alert.signature: Suricata Attack: SQL Injection attempt data.alert.action: allowed data.flow_id: 1431683324388821.000000 data.dest_ip: 192.168.8.128 data.proto: TCP data.http.hostname: 192.168.8.128 data.http.protocol: HTTP/1.1 data.http.http_method: GET data.http.http_content_type: text/html data.http.length: 3725 data.http.uri: /index.php?id=1'%20UNION%20SELECT%201,2,database()--

Рисунок 11 – Результат атаки на Suricata

Snort

t decoder.name	web-accesslog
t full_log	192.168.8.1 - - [31/Mar/2026:15:40:21 -0400] "GET /index.php?id=1'%20UNION%20SELECT%201,2,database()-- HTTP/1.1" 200 3954 "-" "Mozilla/5.0 (Windows NT; Windows NT 10.0; ru-RU) WindowsPowerShell/5.1.19041.6456"
t id	1774986021.16465680
t input.type	log
t location	/var/log/apache2/access.log
t manager.name	debian
t rule.description	Critical Attack: SQL Injection attempt via Apache (Level 12)
# rule.firedtimes	1
t rule.groups	snort_json, web, accesslog
t rule.id	100030
# rule.level	12
rule.mail	true
timestamp	Mar 31, 2026 @ 22:40:21.744

Рисунок 12 – Результат атаки на Snort

Zeek

2026-03-29 00:00 2026-03-29 12:00 2026-03-30 00:00 2026-03-30 12:00 2026-03-31 00:00 2026-03-31 12:00 2026-04-01 00:00		timestamp per 3 hours
t manager.name	debian	
t rule.description	SQL injection attempt.	
# rule.firedtimes	1	
t rule.gdpr	IV_35.7.d	
t rule.groups	web, accesslog, attack, sql_injection	
t rule.id	31103	
# rule.level	7	
rule.mail	false	
t rule.mitre.id	T1190	
t rule.mitre.tactic	Initial Access	
t rule.mitre.technique	Exploit Public-Facing Application	
t rule.nist_800_53	SA.11, SI.4	
t rule.pci_dss	6.5, 11.4, 6.5.1	
t rule.tsc	CC6.6, CC7.1, CC8.1, CC6.1, CC6.8, CC7.2, CC7.3	
timestamp	Mar 31, 2026 @ 22:24:00.020	

Рисунок 13 – Результат атаки на Zeek